

# Practice Questions for Convex Hull

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## 1 Questions

1. In the context of the BN method, what is the primary motivation for transforming a general linear program into a form with only equality constraints and non-negativity restrictions on the variables?
  - a. To simplify the application of the simplex method.
  - b. To facilitate the derivation of the dual problem using the Lagrangian.
  - c. To ensure the feasibility of the primal problem.
  - d. To guarantee the uniqueness of the optimal solution.
  - e. To reduce the computational complexity of the problem.
2. In the BN method, why is the Lagrangian function introduced, and what role does it play in deriving the dual problem?
  - a. The Lagrangian is used to find an initial feasible solution.
  - b. The Lagrangian simplifies the objective function.
  - c. The Lagrangian transforms the constrained primal problem into an unconstrained problem, allowing for easier derivation of the dual.
  - d. The Lagrangian is necessary to ensure the primal problem is convex.
  - e. The Lagrangian is used to prove the strong duality theorem.
3. What is the significance of the weak duality theorem in the context of the BN method, and how does it relate to the optimal solutions of the primal and dual problems?
  - a. Weak duality guarantees the existence of an optimal solution for both the primal and dual problems.
  - b. Weak duality provides a bound on the optimal objective function values of both the primal and dual problems, guiding the search for optimal solutions.
  - c. Weak duality is used to prove the strong duality theorem in the BN method.
  - d. Weak duality is only relevant if the primal problem is infeasible.
  - e. Weak duality is not directly used in the BN method.
4. Explain the concept of a convex hull and how it relates to the feasible region of a linear programming problem.

- a. The convex hull is the smallest convex set containing all points in a given set; in linear programming, the feasible region is always a convex hull.
  - b. The convex hull is the largest convex set contained within a given set; the feasible region of a linear program is always a subset of its convex hull.
  - c. The convex hull is a non-convex set formed by connecting all points in a given set; the feasible region of a linear program is never a convex hull.
  - d. The convex hull is a set of points that are not necessarily connected; the feasible region of a linear program is sometimes a convex hull.
  - e. The convex hull is irrelevant to the feasible region of a linear programming problem.
5. Given the points  $A = (1, 1)$ ,  $B = (3, 1)$ ,  $C = (2, 3)$ , and  $D = (0, 0)$  in  $R^2$ , which of the following points is NOT in the convex hull of  $\{A, B, C, D\}$ ?
- a.  $(1.5, 1)$
  - b.  $(1, 2)$
  - c.  $(2, 2)$
  - d.  $(1, 0.5)$
  - e.  $(2.5, 1.5)$
6. Consider the set of points  $S = \{(0, 0), (1, 0), (0, 1)\}$ . What are the coordinates of a point that lies strictly inside the convex hull of  $S$  but is not a vertex of the convex hull?
- a.  $(0.5, 0.5)$
  - b.  $(1, 1)$
  - c.  $(0, 0)$
  - d.  $(1, 0)$
  - e.  $(0, 1)$
7. Let  $S = \{(x, y) \in R^2 | x^2 + y^2 \leq 4\}$ . Is the convex hull of  $S$  equal to  $S$ ?
- a. Yes
  - b. No
  - c. Only if the origin is included in  $S$
  - d. Only if  $S$  is a convex set
  - e. It depends on the dimension of the space
8. Which of the following sets is NOT convex?
- a. A line segment
  - b. A circle
  - c. A square

- d. A hyperplane
  - e. The set of points  $\{(x, y) | x^2 - y^2 = 1\}$
9. What is the fundamental characteristic of a convex set, and how does this property relate to the feasible region in linear programming problems?
- a. A convex set is a set where any line segment connecting two points within the set is entirely contained within the set. This is crucial in linear programming because the feasible region is always a convex set, ensuring that any convex combination of feasible solutions is also feasible.
  - b. A convex set is a set that can be enclosed by a circle. This is important in linear programming because it simplifies the search for an optimal solution.
  - c. A convex set is a set with only straight lines as boundaries. This is relevant in linear programming because linear constraints define straight lines.
  - d. A convex set is a set that is always connected. This is important in linear programming because the feasible region must be a connected set.
  - e. A convex set is a set that contains its boundary. This is relevant in linear programming because the boundary of the feasible region is part of the feasible region.
10. Given the points  $z_1 = (1, 2)$  and  $z_2 = (3, 4)$  in  $R^2$ , what is a convex combination of  $z_1$  and  $z_2$  where the resulting point has an x-coordinate of 2?
- a.  $2z_1 + 0z_2$
  - b.  $0z_1 + 2z_2$
  - c.  $0.5z_1 + 0.5z_2$
  - d.  $0.25z_1 + 0.75z_2$
  - e.  $0.75z_1 + 0.25z_2$

## 2 Answers

1. In the context of the BN method, what is the primary motivation for transforming a general linear program into a form with only equality constraints and non-negativity restrictions on the variables?
  - a. While the simplex method can be applied, this isn't the primary motivation of the transformation in the BN method.
  - b. The BN method uses the Lagrangian to derive the dual. Having only equality constraints simplifies this process significantly.
  - c. The transformation doesn't inherently guarantee feasibility; feasibility is a separate concern.
  - d. The transformation doesn't guarantee uniqueness; multiple optimal solutions can still exist.
  - e. The transformation might reduce complexity in some cases, but it's not the primary goal of the BN method.
2. In the BN method, why is the Lagrangian function introduced, and what role does it play in deriving the dual problem?
  - a. Finding an initial feasible solution is a separate concern, not the role of the Lagrangian in the BN method.
  - b. The Lagrangian doesn't simplify the objective function; it incorporates the constraints into the objective function.
  - c. The Lagrangian is the key tool in the BN method. By removing the constraints, it allows for a straightforward derivation of the dual problem.
  - d. Convexity is important in linear programming, but the Lagrangian's role in the BN method is not directly related to ensuring convexity.
  - e. The Lagrangian is used in the derivation of the dual, but it doesn't directly prove strong duality.
3. What is the significance of the weak duality theorem in the context of the BN method, and how does it relate to the optimal solutions of the primal and dual problems?
  - a. Weak duality doesn't guarantee the existence of optimal solutions; other conditions are needed.
  - b. Weak duality provides an upper bound for the primal and a lower bound for the dual, crucial for the BN method's convergence.
  - c. While related, weak duality is a separate theorem; strong duality is a consequence of weak duality under certain conditions.
  - d. Weak duality applies regardless of the feasibility of the primal problem.
  - e. Weak duality is fundamental to the BN method's logic and convergence.
4. Explain the concept of a convex hull and how it relates to the feasible region of a linear programming problem.
  - a. The convex hull of a set of points is the smallest convex set that contains all the points. In linear programming, the feasible region, defined by the intersection of linear constraints, is always a convex set. Therefore, the feasible region is either itself a convex hull (if it's a polygon) or is contained within a larger convex hull that includes points outside the feasible region.

- b. The convex hull is the \*smallest\*, not largest, convex set containing all points. The feasible region is a subset of its convex hull only if the feasible region is not itself a convex hull (e.g., if it's a bounded region).
  - c. A convex hull is, by definition, a convex set. The feasible region of a linear program is always a convex set.
  - d. The points in a convex hull are connected to form a convex set. The feasible region of a linear program is always a convex set.
  - e. The convex hull is fundamental to understanding the feasible region in linear programming because the feasible region is always a convex set.
5. Given the points  $A = (1, 1)$ ,  $B = (3, 1)$ ,  $C = (2, 3)$ , and  $D = (0, 0)$  in  $R^2$ , which of the following points is NOT in the convex hull of  $\{A, B, C, D\}$ ?
  - a.  $(1.5, 1)$  is the midpoint of AB, which is clearly in the convex hull.
  - b. The point  $(1, 2)$  is not in the convex hull. The convex hull is the smallest convex set containing all the points. While  $(1, 2)$  is above the line segment AC, it is not within the polygon formed by A, B, C, and D.
  - c.  $(2, 2)$  is inside the quadrilateral ABCD, thus within the convex hull.
  - d.  $(1, 0.5)$  lies on the line segment AD, and is therefore in the convex hull.
  - e.  $(2.5, 1.5)$  lies inside the quadrilateral ABCD, thus within the convex hull.
6. Consider the set of points  $S = \{(0, 0), (1, 0), (0, 1)\}$ . What are the coordinates of a point that lies strictly inside the convex hull of  $S$  but is not a vertex of the convex hull?
  - a. The point  $(0.5, 0.5)$  is a convex combination of  $(0, 0)$ ,  $(1, 0)$ , and  $(0, 1)$ , specifically  $(0.5, 0.5) = 0.5(1, 0) + 0.5(0, 1)$ . It lies inside the triangle formed by the three points.
  - b.  $(1, 1)$  is outside the triangle formed by the points in  $S$ .
  - c.  $(0, 0)$  is a vertex of the convex hull.
  - d.  $(1, 0)$  is a vertex of the convex hull.
  - e.  $(0, 1)$  is a vertex of the convex hull.
7. Let  $S = \{(x, y) \in R^2 | x^2 + y^2 \leq 4\}$ . Is the convex hull of  $S$  equal to  $S$ ?
  - a. Yes, because  $S$  is already a convex set (it's a disk). The convex hull of a convex set is the set itself.
  - b. This is incorrect because  $S$  is a convex set.
  - c. The convex hull is independent of whether the origin is included.
  - d. The question is about whether the convex hull of  $S$  is equal to  $S$ . The fact that  $S$  is convex is sufficient to answer the question.
  - e. The dimension of the space is irrelevant since  $S$  is already defined in  $R^2$ .
8. Which of the following sets is NOT convex?
  - a. A line segment is a convex set.

- b. A circle is a convex set.
  - c. A square is a convex set.
  - d. A hyperplane is a convex set.
  - e. The set of points defined by  $x^2 - y^2 = 1$  is a hyperbola, which is not a convex set. A line segment connecting two points on the hyperbola will not be entirely contained within the hyperbola.
9. What is the fundamental characteristic of a convex set, and how does this property relate to the feasible region in linear programming problems?
- a. A convex set is defined as a set where any line segment connecting two points within the set is entirely contained within the set. This property is fundamental to linear programming because the feasible region of a linear program is always a convex set. This convexity ensures that any weighted average (convex combination) of feasible solutions is also a feasible solution. This property is crucial for the development and analysis of algorithms that solve linear programming problems, as they often rely on the convexity of the feasible region to guarantee convergence to an optimal solution.
  - b. A convex set is not necessarily a set that can be enclosed by a circle. Consider a convex polygon; it is a convex set but cannot always be enclosed by a circle. The shape of the feasible region is not restricted to being circular.
  - c. A convex set can have curved boundaries. For example, a circle is a convex set, but its boundary is a curve. The feasible region can have curved boundaries in non-linear programming problems, but in linear programming, the boundaries are straight lines.
  - d. A convex set is not necessarily connected. A disconnected set can still be convex if each of its connected components is convex. The feasible region in linear programming is always a connected set.
  - e. A convex set does contain its boundary, but this is not the defining characteristic of a convex set. The defining characteristic is that the line segment between any two points in the set is also in the set.
10. Given the points  $z_1 = (1, 2)$  and  $z_2 = (3, 4)$  in  $R^2$ , what is a convex combination of  $z_1$  and  $z_2$  where the resulting point has an x-coordinate of 2?
- a. This is not a convex combination because it does not use  $z_2$ . A convex combination must use all points in the set.
  - b. This is not a convex combination because it does not use  $z_1$ . A convex combination must use all points in the set.
  - c. A convex combination is of the form  $tz_1 + (1-t)z_2$  where  $0 \leq t \leq 1$ . If the x-coordinate is 2, then  $t(1) + (1-t)(3) = 2$ , which simplifies to  $t + 3 - 3t = 2$ , so  $-2t = -1$ , and  $t = 0.5$ . Therefore, the convex combination is  $0.5z_1 + 0.5z_2$ .
  - d. This gives an x-coordinate of  $0.25(1) + 0.75(3) = 2.5$ , not 2.
  - e. This gives an x-coordinate of  $0.75(1) + 0.25(3) = 1.5$ , not 2.